A. Triangle

time limit per test: 2 seconds
memory limit per test: 64 megabytes

At a geometry lesson Bob learnt that a triangle is called right-angled if it is nondegenerate and one of its angles is right. Bob decided to draw such a triangle immediately: on a sheet of paper he drew three points with integer coordinates, and joined them with segments of straight lines, then he showed the triangle to Peter. Peter said that Bob's triangle is not right-angled, but is almost right-angled: the triangle itself is not right-angled, but it is possible to move one of the points exactly by distance 1 so, that all the coordinates remain integer, and the triangle become right-angled. Bob asks you to help him and find out if Peter tricks him. By the given coordinates of the triangle you should find out if it is right-angled, almost right-angled, or neither of these.

Input

The first input line contains 6 space-separated integers $x_1, y_1, x_2, y_2, x_3, y_3$ — coordinates of the triangle's vertices. All the coordinates are integer and don't exceed 100 in absolute value. It's guaranteed that the triangle is nondegenerate, i.e. its total area is not zero.

Output

If the given triangle is right-angled, output `RIGHT`, if it is almost right-angled, output `ALMOST`, and if it is neither of these, output `NEITHER`.

Examples

input Copy

```
0 0 2 0 0 1
```

output Copy

```
RIGHT
```

input Copy

```
2 3 4 5 6 6
```

output Copy

```
NEITHER
```

input Copy

```
-1 0 2 0 0 1
```

output Copy

```
ALMOST
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 18 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370628041	Apr/11/2026 09:13	Accepted
370627703	Apr/11/2026 09:09	Accepted
370627441	Apr/11/2026 09:06	Compilation error

370627208	Apr/11/2026 09:04	Wrong answer on test 31
370626934	Apr/11/2026 09:01	Compilation error

→ Problem tags

brute force | geometry | *1500

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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